

Title: Identification of Statistically Significant Factors Contributing to Severity of Car Accidents

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Abstract

Due to increasing number of ways to analyze data, there are additional methods that can be employed to identify contributing factors based on the patterns of the data. Some of these methods include machine learning algorithms and statistical models, which can be performed to identify such factors in an efficient fashion. The crash data involved in the study were obtained from Highway Safety Information System (HSIS) to cover 5-year time period and multiple crash components containing driver, roadway, vehicle and other environmental information. The statistical modeling were implemented via the software, R, to develop two generalized linear models which include multinomial and binary logit regression, applied to dichotomous and multiple divisions of the crash severity, respectively. The results of both models are somewhat similar, leading to several probabilities between several factors that lead to various types of collisions. Specially, it is concluded with 95% confidence level that the features that have been identified through the models are the level of severity, the light conditions, as well as the environment in which the collisions take place.

Introduction

Traffic accidents and the different circumstances within each incident generates forms of data that is able to be analyzed. By translating this information, the least critical factors in determining a hit and run that would occur could be filtered out. This would provide useful information to help reduce traffic and produce safer roadways and vehicles. Rendering this information is the task of a data scientist. Their role, specifically those majoring in civil engineering, is to transform unruly information into an easier to comprehend and usable format.

Materials and Methods

The general method used in this research project was that of machine learning, which is applying a statistical format to interpret results so that our program is able to improve its own performance on interpreting the data we provide. An application of regression analysis was also used in order to determine contrasting relationships between each individual data.

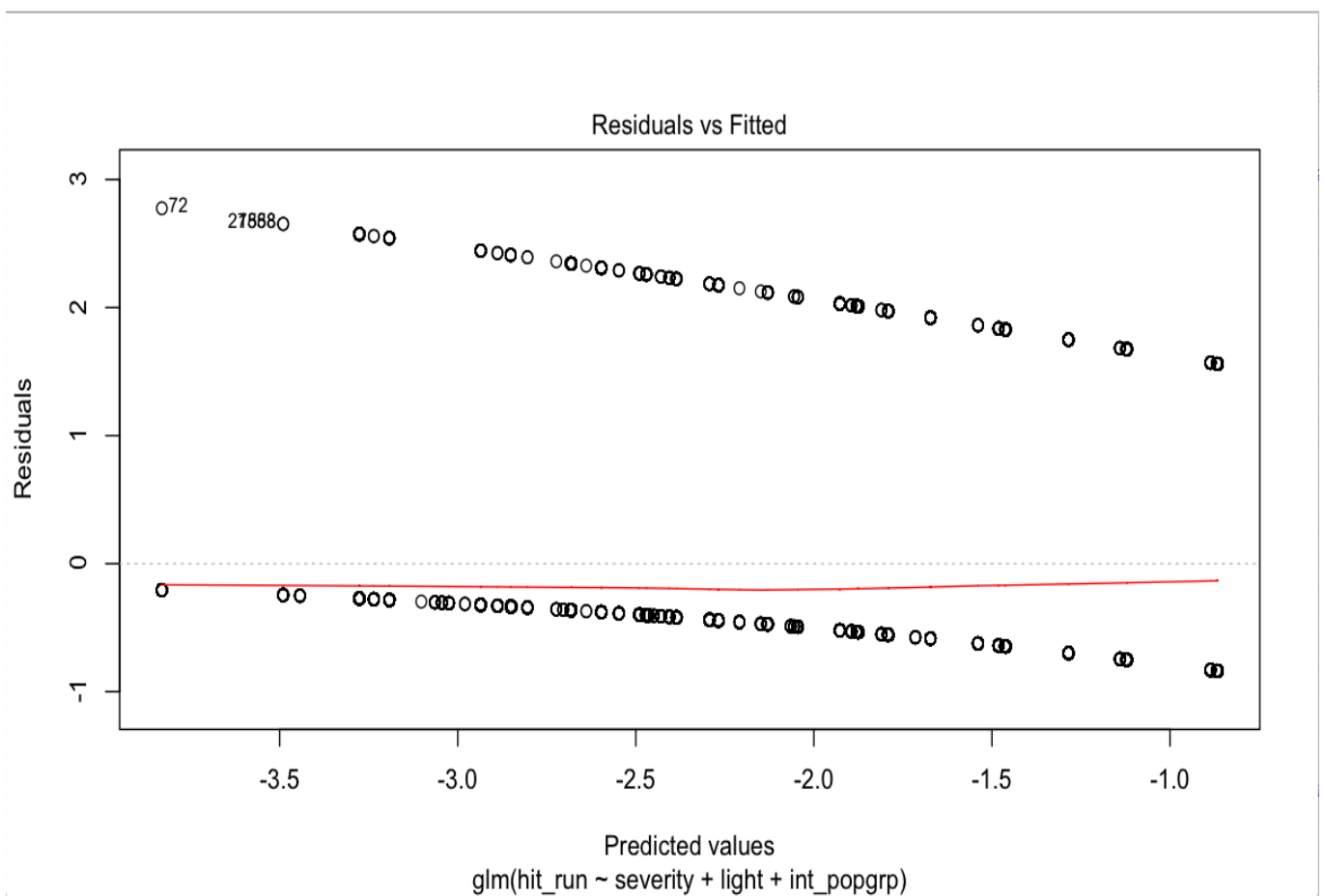
This study uses the Highway Safety Information Systems as a database over a 5-year time period. It contributes the information and details about all of the factors. These variables include rodwycls, weekday, acctype, hwy_grp, hour, pop_grp, weather1, severity, hit_run, ped_actn, rdsurf, rd_defl, light, numbehs, typedesc, lght_typ, ml_mast, ml_left, ml_right, ml_trflo, xstrtmst, xstrtlft, xstrtrgh, xsttrflo, trf_cntl, hwygrp.1, int_popgrp, ml_aadt, xstaadt, ml_lanes, and xstlanes. Which are translated to, in order, roadway classification, weekday, type-of-collision, highway group, hour, population group, weather, hit and run, pedestrian action, road surface, mainlines, intersection streets, and traffic control. Within each variable were categories or values on how each piece of data is classified. This was represented by an alphabetical or numerical value.

To run this data through the R software, we had to translate certain factors into either binomial or multinomial values. This includes clumping data and representing them of values to 1 or 2 for binomial and 1, 2, or 3 for multinomial. For example, severity of a car accident is defined with '1' being fatal, '2' severe injury, '3' other visible injury, '4' complaint of pain' and '0' property damage only. We would group being fatal and severe injury as '1' while band the others as '2' to form a binomial model.

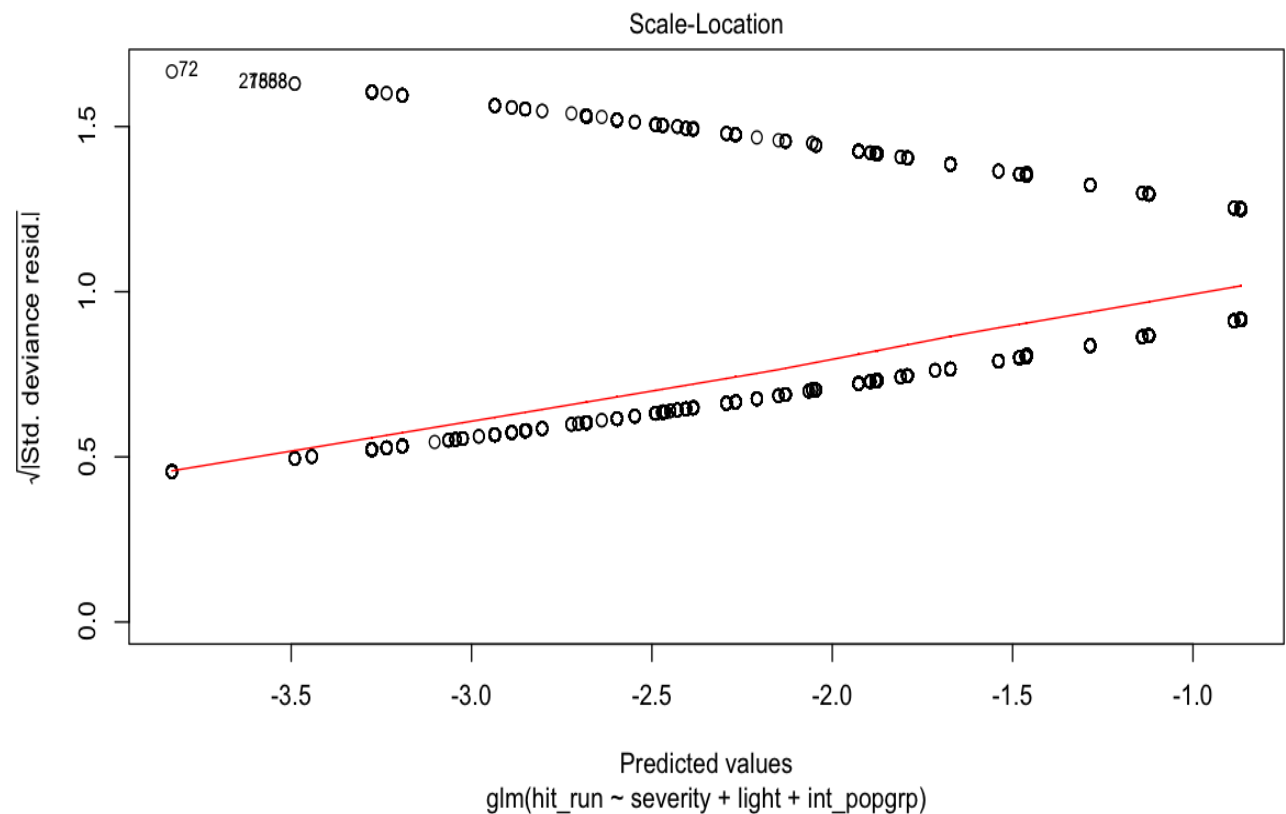
The use logistic regression from statistics to analyze the data after the translations. The dependent variable was hit and run because it is dichotomous in nature. This is then compared to every other factor and resulted in different correlations. The further the correlation had to the hit and run factor, the more likely it is to be removed. This method is called backwards deletion and is only concluded until the model is left with only significant terms.

Results

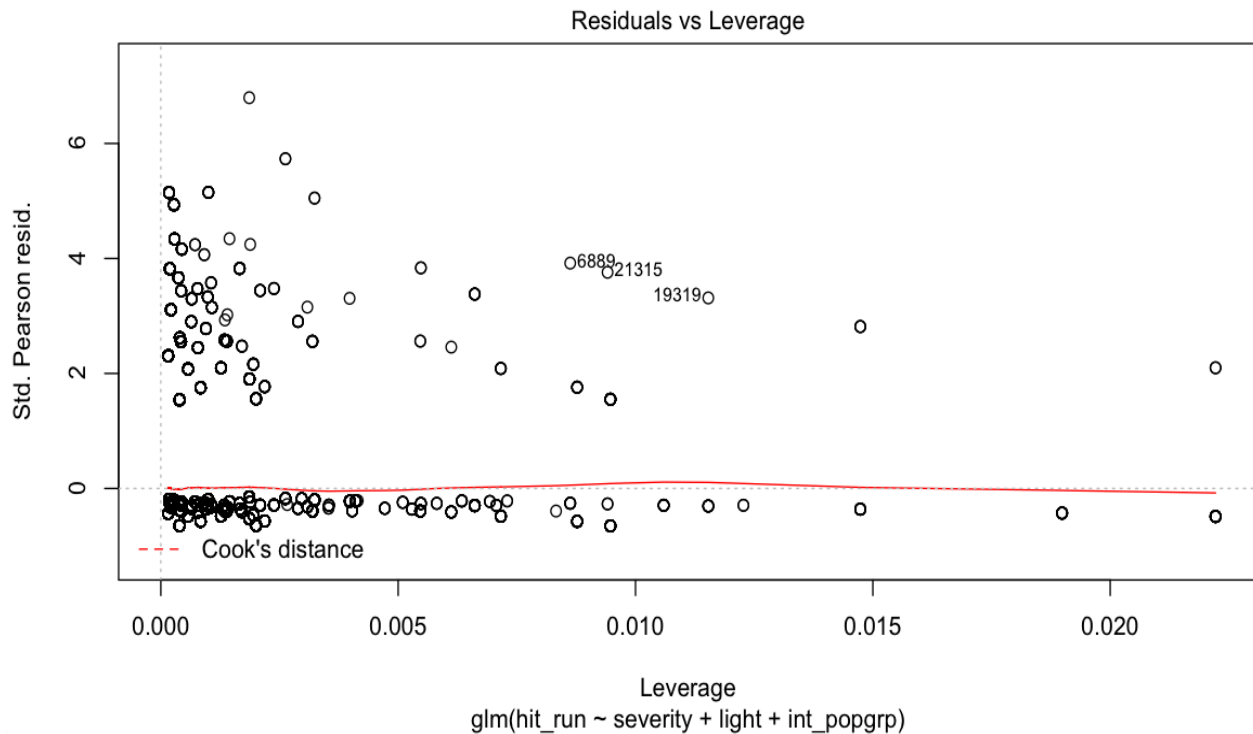
The study resulted in a generalized linear model for hit and run with the variables severity, light, and intersection population. This concludes that depending on if those factors are present in an incident, there exists a higher chance for a driver to do a hit and run. These terms are considered as residuals.



Residuals vs Fitted shows if the residuals have non-linear patterns. There exist spread residuals around a horizontal line without any distinct pattern. This indicates that there is not a non-linear relationship.



Scale location determines whether the residuals are spread equally along the predictor's range. The line starts to incline upwards because the data points start to incline upwards as well.



The residuals vs leverage plot guides in determining influential data points to the model. The outliers against the red line are not necessarily influential. The points with values are considered to be the most influential, and removing them would alter the regression.

Discussion

Analyzing factors of traffic occurrences is useful to society because it would provide a basis to advance the technology. This in turn would hopefully make traffic accidents less frequent for the future. In this specific study, we looked at the most influential factors that existed to cause a driver to undertake a hit and run.

From the Highway Safety Information Systems guidebook, we were able to obtain data throughout a five-year timespan. This data included information about drivers, roadways, vehicles, and other environmental information. This data was then implemented via the software, R. We transformed the hit and run data into a binary logit regression. Next, we removed factors based on the Akaike Information Criterion (AIC). This is the maximum likelihood estimate that would help to prevent overfitting on our graphs.

Conclusion

The study concludes with the most important factors in relation with the hit and run variable. These include the severity of the accident, whether or not there was light in the scene, and how many other drivers were witness at the intersection.

References

- Gardener, M. (2012). *Beginning R: The statistical programming language*. Indianapolis: John Wiley & Sons.
- Nujjetty, A. P., & Mohamedshah, Y. M. (2014). *Highway Safety Information System*. Washington, D.C.: U.S. Dept. of Transportation, Federal Highway Administration.